SUBASH BALAJI S

Generative AI & Backend Engineer

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PROFILE SUMMARY

Engineering student building **LLM-powered multi-agent systems**, **RAG applications**, and **production-ready REST APIs** with Python, FastAPI, and LangChain. Experienced in deploying containerized services with Docker and CI/CD. Seeking a software engineering internship to ship scalable, AI-driven backend products.

PROJECTS

Pagetrail: Document Q&A Assistant *Python, FastAPI, LangGraph, Gemini, Supabase PostgreSQL, React, TypeScript, Docker*

- Built and deployed on Render a full-stack, document-grounded AI assistant with a **FastAPI** backend and **React + TypeScript** frontend, using **LangGraph** to orchestrate Gemini with Tavily web search and long-term memory tools.
- Implemented a **RAG pipeline** that parses PDF, DOCX, TXT, and Markdown uploads, splits them into chunks, generates **embeddings** stored in **Supabase PostgreSQL**, and retrieves the most relevant chunks per conversation using **cosine similarity**.
- Designed **persistent storage** on **Supabase PostgreSQL** with SQLAlchemy for users, conversations, messages, long-term memory, and document embeddings; added **JWT authentication**, row-level security, and **Docker Compose** packaging.

Multi-Agent Research System *Python, LangChain, Gemini, Groq, Tavily, Streamlit, Trafilatura*

- Built a modular **multi-agent research pipeline** combining autonomous web search, LLM report synthesis (**Gemini + Groq**), and a critical-review agent, cutting manual research effort by an estimated **80%**.
- Orchestrated **4+ AI tools** through LangChain's agent framework into a unified Streamlit app that exports structured Markdown reports, using prompt engineering to control agent reasoning and output format.

LangChain AI Assistant *Python, LangChain, Google Gemini, Tavily, Weatherstack, Streamlit*

- Developed a conversational AI assistant with **persistent session memory** for context-aware multi-turn dialogue; deployed on **Render** with a Streamlit interface.
- Integrated **2 real-time tools** (Tavily web search, Weatherstack weather lookup) as LangChain tools, with structured error handling and graceful fallbacks when API keys are missing.

Products Management REST API *FastAPI, PostgreSQL, SQLAlchemy, Pydantic, Alembic, Docker, GitHub Actions*

- Designed a **RESTful API** with full CRUD, **Pydantic v2** validation, query filtering, **JWT authentication**, **role-based access control (RBAC)**, and standardized HTTP status codes with structured error handling.
- Configured **PostgreSQL** with SQLAlchemy ORM and Alembic migrations, published auto-generated Swagger/OpenAPI documentation, and automated deployment using **Docker Compose** and **GitHub Actions** CI/CD.

TECHNICAL SKILLS

Languages: Python (primary), JavaScript (ES6+), TypeScript, SQL, C, C++, Bash
Frontend: React, TypeScript, Vite, HTML5, CSS3, Streamlit, responsive design
Backend / API: FastAPI, REST API design, Pydantic v2, JWT, OAuth2, RBAC, middleware, dependency injection, async/await, Uvicorn
Database: PostgreSQL, Supabase, SQLAlchemy ORM, Alembic, Redis (cache & broker), data persistence, schema design, indexing
Jobs & Files: Celery, async task queues, PDF parsing (pdfplumber, PyMuPDF), MinIO/S3, email notifications
DevOps / Cloud: Docker, Docker Compose, GitHub Actions, CI/CD, Render, Railway, DigitalOcean, AWS (EC2, S3, IAM, VPC, RDS, Route 53)
AI / LLM: LangChain, LangGraph, Gemini, Groq, OpenAI API, Tavily, RAG, embeddings, semantic search, prompt engineering, multi-agent systems
Testing & Practices: pytest, httpx, Postman, Swagger/OpenAPI, Git, logging, bcrypt, system design

EDUCATION

Dayananda Sagar College of Engineering Bengaluru, Karnataka
Bachelor of Engineering in Medical Electronics Engineering *2024 – 2028*
GPA: 8.0/10 | Editor, Department Newsletter

CERTIFICATIONS & ACHIEVEMENTS

- **AWS Solutions Architect – Fundamentals** (Amazon Web Services, 2024) — Studied EC2, S3, IAM, VPC, and RDS.
- **CS50: Introduction to Programming with Python** (Harvard University)
- **Data Structures & Algorithms Specialization** (UC San Diego) — In progress.
- **Programming with C** and **Working with Excel** (Infosys Springboard, 2024)